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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations
December 2025

AD2005-1 - DATA PRIVACY AND INTERNET SECURITY
(For AID)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- 1 **Cryptography is primarily concerned with:**
A) Securing data from unauthorized access B) Increasing computation speed
C) Data compression D) Data redundancy
- 2 **Brute-force attack in cryptography means:**
A) Using mathematical proofs to break a cipher B) Trying all possible key combinations
C) Exploiting software bugs D) Stealing the key physically
- 3 **Caesar Cipher is an example of:**
A) Transposition cipher B) Monoalphabetic substitution cipher
C) Polyalphabetic substitution cipher D) One-time pad
- 4 **Mono-alphabetic ciphers are vulnerable to:**
A) Brute-force attack B) Frequency analysis
C) Timing attack D) Linear cryptanalysis
- 5 **Which cipher uses a 5×5 matrix of letters for encryption?**
A) Hill Cipher B) Playfair Cipher
C) Vigenère Cipher D) One-time Pad
- 6 **Hill cipher encryption is based on:**
A) Modular arithmetic and matrix multiplication B) Frequency analysis
C) Stream ciphers D) Bitwise XOR operations
- 7 **The One-Time Pad (OTP) is:**
A) Unconditionally secure B) Conditionally secure
C) Insecure D) Deprecated
- 8 **Block ciphers encrypt:**
A) 1 bit at a time B) Multiple bits at once in a block
C) Only ASCII text D) Random numbers
- 9 **The block size of DES is:**
A) 32 bits B) 56 bits
C) 64 bits D) 128 bits
- 10 **The DES key length is:**
A) 56 bits (effective key) B) 64 bits (with parity)
C) 56 bits and 64 bits D) 128 bits



- 11 Feistel cipher structure divides the block into:
 A) 2 halves (Left and Right) B) 4 quarters
 C) 8 sub-blocks D) Only 1 block
- 12 In a public-key cryptosystem, each user has:
 A) Only one key B) A pair of keys (public and private)
 C) A secret shared key D) No keys
- 13 Which is NOT a requirement of a public-key cryptosystem?
 A) Easy to generate key pairs B) Easy to compute ciphertext from plaintext
 C) Easy to compute plaintext from ciphertext without private key D) Computationally infeasible to compute private key from public key
- 14 Applications of public-key cryptography include:
 A) Encryption B) Digital signatures
 C) Key exchange D) All of the above
- 15 RSA security is based on the difficulty of:
 A) Discrete logarithm problem B) Integer factorization problem
 C) Modular addition problem D) Elliptic curve arithmetic
- 16 In Diffie-Hellman, both parties agree upon:
 A) A large prime number (p) and a primitive root (g) B) Only one secret number
 C) A private key only D) A ciphertext block
- 17 Which attack is Diffie-Hellman vulnerable to?
 A) Replay attack B) Man-in-the-middle attack
 C) Brute force D) Frequency analysis
- 18 El Gamal encryption is based on:
 A) Integer factorization B) Discrete logarithm problem
 C) Elliptic curve arithmetic D) Matrix multiplication
- 19 The ciphertext in El Gamal consists of:
 A) One integer B) Two integers (c1, c2)
 C) Three numbers D) Hash value
- 20 Elliptic Curve Cryptography is based on:
 A) Integer factorization B) Discrete logarithm problem on elliptic curves
 C) Modular division D) Key stretching

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Classify and explain different types of cryptanalytic attacks on encrypted messages with appropriate examples	08	L*2	1	1,2
	b) Demonstrate the process of encryption and decryption in the Hill Cipher with an example, and explain the factors influencing its security.	08	L2	1	1,2
2.	a) Using a neat diagram, illustrate and explain the working of DES encryption and decryption processes, showing how each stage transforms the data	08	L2	1	1,2
	b) Using a neat diagram, demonstrate and explain how a public key cryptosystem provides both authentication and secrecy	08	L2	1	1,2

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| 3. a) | Explain the difference between cryptography, Cryptanalysis, and Brute – Force attack. | 08 | L2 | 3 | 1,3 |
| b) | Using the Playfair cipher and the key 'MONARCHY,' apply the encryption process to the plaintext 'attack,' showing each step of the matrix formation and substitution. | 08 | L3 | 1 | 1,2 |

Unit – II

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|-------|--|----|----|---|-----|
| 4. a) | Using the Diffie-Hellman Key Exchange method, apply the given parameters ($q = 17$, primitive root = 5, Alice's secret key = 4, Bob's secret key = 6) to compute the common secret key exchanged between Alice and Bob | 08 | L3 | 2 | 1,2 |
| b) | Apply the principles of Elliptic Curve Cryptography to demonstrate, with a worked-out example, the process of encryption and decryption of a message | 08 | L3 | 2 | 1,2 |
| 5. a) | Explain how public key certificates support secure communication, and describe the structure and key components of an X.509 certificate. | 08 | L2 | 3 | 2,3 |
| b) | Apply the RSA algorithm to encrypt and decrypt a given message by selecting appropriate prime numbers and Keys, and demonstrate the complete process with Calculations. | 08 | L3 | 4 | 1,2 |
| 6. a) | Explain the EL Gamal cryptographic system with encryption and decryption steps. | 08 | L2 | 3 | 2,3 |
| b) | Apply the principles of Diffie – Hellman Key exchange method to identify potential challenges in Key – exchange between two end users. | 08 | L3 | 4 | 1,2 |

Unit – III

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|-------|--|----|----|---|-----|
| 7. a) | Explain the Randomization Method in privacy-preserving data mining. Illustrate with an example how sensitive data can be protected while still enabling useful data analysis. Discuss its advantages and limitations. | 08 | L2 | 4 | 2,3 |
| b) | Explain the concept of Distributed Privacy-Preserving Data Mining. Demonstrate, with an example, how data mining can be performed across multiple sites without sharing sensitive individual data. Highlight the challenges in distributed privacy preservation. | 08 | L2 | 4 | 2,3 |
| 8. a) | Describe the various applications of privacy-preserving data mining in real-world scenarios such as healthcare, finance, or e-commerce. Choose one application and explain in detail how privacy is maintained while deriving useful insights. | 08 | L2 | 5 | 1,2 |
| b) | Discuss Group-Based Anonymization techniques. Describe how k-anonymity, l-diversity, or t-closeness work to protect individual records. Provide a detailed example showing anonymization of a sample dataset. | 08 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
